

State space Analysis

1.	A state space model has 3 state variables. How many first-order differential equations describe the system? A. 1 B. 2 C. 3 D. 6
2.	If a system has 3 state variables, 2 inputs, and 1 output, then the dimensions of matrices A, B, C, D are respectively: A. $3 \times 3, 3 \times 2, 1 \times 3, 1 \times 2$ B. $3 \times 3, 2 \times 3, 3 \times 1, 2 \times 1$ C. $2 \times 2, 3 \times 2, 1 \times 3, 3 \times 1$ D. $3 \times 2, 2 \times 3, 3 \times 1, 1 \times 2$
3.	The characteristic equation of a state matrix A is obtained by: A. $A = 0$ B. $sI + A = 0$ C. $sI - A = 0$ D. $I - sA = 0$
4.	The state transition matrix $\Phi(t)$ is defined as: A. e^{At} B. At C. $I + At$ D. $A^{-1}t$
5.	If $A = 0$, then the state transition matrix $\Phi(t)$ is: A. 0 B. At C. I D. e^t

6.	If the state transition matrix is $\Phi(t) = e^{At}$, then $\Phi(0)$ is: A. 0 B. A C. I D. e^A
7.	The solution of the state equation $\dot{x} = Ax$ with initial condition $x(0)$ is: A. $x(t) = Ax(0)$ B. $x(t) = e^{At}x(0)$ C. $x(t) = A^t x(0)$ D. $x(t) = e^t Ax(0)$
8.	A series RLC circuit has the equation: $L \frac{di}{dt} + Ri + \frac{1}{C} \int i dt = v(t)$ Choose state variables and write state equations.
9.	Given the differential equation: $\ddot{y} + 5\dot{y} + 6y = u(t)$ Convert to state-space form using $x_1 = y, x_2 = \dot{y}$.
10.	Equation: $\ddot{x} + 6\dot{x} + 11x + 6x = 0$ Find state variables and write A matrix.

11.	Given: $G(s) = \frac{Y(s)}{U(s)} = \frac{3}{s^2 + 5s + 6}$ Tasks: 1. Draw the Signal Flow Graph (SFG) 2. Find the state-space model (controllable canonical form)
12.	Problem: $G(s) = \frac{2s + 5}{s^2 + 4s + 3}$ Draw SFG and form state-space model.
13.	$G(s) = \frac{4}{s^3 + 6s^2 + 11s + 6}$ Obtain the state equation
14.	A series RLC circuit is given: • Resistor: $R = 2\ \Omega$ • Inductor: $L = 1\ H$ • Capacitor: $C = 0.5\ F$ Voltage source: $v(t)$ applied across the series elements. Task: 1. Draw the Signal Flow Graph (SFG). 2. Find the state-space representation.

15.	• $R = 1\ \Omega$, $L = 1\ \text{H}$, $C = 1\ \text{F}$ • Parallel combination with voltage source $V_s(t)$ across the network • Find SFG and state-space model for capacitor voltage output.
16.	A control system is represented by $\dot{x} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} x$ How many state variables does the system have? A. 1 B. 2 C. 3 D. 4
17.	The transfer function of a system is $G(s) = \frac{5}{s^2 + 4s + 3}$ What is the minimum number of state variables required? A. 1 B. 2 C. 3 D. 5